

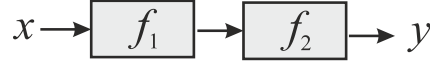
Реализовать распараллеливание на основе конвейерной модели вычислений.

Сравнить эффективность последовательной и параллельной версий программ.

Реализовать распараллеливание а) с использованием блоков сообщений и программных агентов; б) с использованием блока сообщений **concurrency::transformer**

$$f_1 : \mathbb{R} \rightarrow \mathbb{R}, f_2 : \mathbb{R} \rightarrow \mathbb{R};$$

$$f = f_2 \circ f_1 : \mathbb{R} \rightarrow \mathbb{R}, y = f(x)$$



$$1. \quad f_1(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+n}{1+|x|+n^3+(k^2+j^2)^{3/2}}$$

$$f_2(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x}{1+|x|+n^2+k^3+j^2}$$

$$2. \quad f_1(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+n}{1+|x|+n^3+k^2+j^2}$$

$$f_2(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x}{1+|x|+(n^2+j^2)^{3/2}+k^3}$$

$$3. \quad f_1(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+j}{1+\sqrt{x^2+n^4+k^2+j^3}}$$

$$f_2(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x-k}{1+|x|+n^3+k^4+j^2}$$

$$4. \quad f_1(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2+n}{1+x^2+n^3+(k^2+j^2)^{3/2}}$$

$$f_2(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2}{1+x^2+n^2+k^3+j^2}$$

$$5. \quad f_1(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2+n}{1+x^2+n^3+k^2+j^2}$$

$$f_2(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2}{1+x^2+(n^2+j^2)^{3/2}+k^3}$$

$$6. \quad f_1(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+j}{1+\sqrt{x^2+n^4+k^2+j^3}}$$

$$f_2(x) = \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2-k}{1+x^2+n^3+k^3+j^2}$$

$$\begin{aligned}
7. \quad f_1(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{\sqrt{|x|+n}}{\sqrt{1+|x|+n^5+(k^2+j^2)^{3/2}}} \\
f_2(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{\sqrt{|x|}}{\sqrt{1+|x|+n^2+k^3+j^2}} \\
8. \quad f_1(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{\sqrt{|x|+j}}{\sqrt{1+j^5+\sqrt{|x|+n^4+k^2}}} \\
f_2(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x-k}{1+|x|+n^3+k^4+j^2} \\
9. \quad f_1(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+n}{1+|x|+n^3+(k^2+j^2)^{3/2}} \\
f_2(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+\ln(k+1)}{1+|x|+n^2+k^3+j^2} \\
10. \quad f_1(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2+n}{1+x^2+n^3+k^2\ln(1+k)+j^2} \\
f_2(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2}{1+x^2+(n^2+j^2)^{3/2}+k^3} \\
11. \quad f_1(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2+n\ln(1+n)}{1+x^2+n^3+(k^2+j^2)^{3/2}} \\
f_2(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x^2}{1+x^2+n^2+k^3+j^2} \\
12. \quad f_1(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x+\ln(j+1)}{1+\sqrt{x^2+n^4+k^2}+j^3} \\
f_2(x) &= \sum_{n=0}^N \sum_{k=0}^N \sum_{j=0}^N \frac{x-k\ln(k+1)}{1+|x|+n^3+k^4+j^2}
\end{aligned}$$

$$14. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}}} \right)$$

$$15. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2 y + e^{\sqrt{k}}} \right)$$

$$16. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\cos^2 y + e^{\sqrt{k}}} \right)$$

$$17. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2(x + y) + e^{\sqrt{k}}} \right)$$

$$18. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\cos^2(x + y) + e^{\sqrt{k}}} \right)$$

$$19. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{y^2 + x^2 + e^{\sqrt{k}} \cos \frac{1}{k+1}} \right)$$

$$20. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}} \cos \frac{1}{k+1}} \right)$$

$$21. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}} \cos \frac{1}{k+1}} \right)$$

$$22. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2 y + e^{\sqrt{k}} \cos \frac{1}{k+1}} \right)$$

$$23. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{y^2 + x^2 + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$

$$24. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$

$$25. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$

$$26. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2 y + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$